

Introduction to Online Convex Optimization

Reading Seminar — Sections 5.5–5.8

Rintaro Okahara

2026-06-16

1. Randomized Regularization	2
2. Adaptive Gradient Descent (AdaGrad)	11
3. Bibliographic Remarks	27
4. Exercises	29

1. Randomized Regularization

1.1 Follow-the-Perturbed-Leader

Algorithm 16 — Follow-the-perturbed-leader for convex losses

- 1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
- 2: Let $x_1 = \mathbb{E}_{n \sim \mathcal{D}}[\arg \min_{x \in \mathcal{K}} \{n^\top x\}]$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict x_t .
- 5: Observe f_t , suffer $f_t(x_t)$, and set $\nabla_t = \nabla f_t(x_t)$.
- 6: Update

$$x_{t+1} = \mathbb{E}_{n \sim \mathcal{D}} \left[\arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^t \nabla_s^\top x + n^\top x \right\} \right] \quad (5.4)$$

7: **end for**

- Resembles a **special case of FTRL** (Follow-the-Regularized-Leader).
- **Key difference:** the regularizer is **randomized** — noise $n \sim \mathcal{D}$ is injected into the objective in place of a fixed regularizer.

1.2 Regret Bound for FPL

Theorem (5.8)

Let $\mathcal{D}$ be (σ, L) -stable with respect to the norm $\|\cdot\|_a$. Then FPL (Algorithm 16) attains

$$\text{Regret}_T \leq \eta D (G^*)^2 L T + \frac{1}{\eta} \sigma D,$$

and, optimizing over η ,

$$\text{Regret}_T \leq 2LDG^* \sqrt{\sigma T}.$$

Definition $((\sigma, L)$ -stability)

For $\sigma, L \in \mathbb{R}$, a distribution $\mathcal{D}$ is (σ, L) -stable with respect to $\|\cdot\|_a$ if

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \forall u : \int_n |\mathcal{D}(n) - \mathcal{D}(n - u)| \, dn \leq L \|u\|_a^*.$$

- ...
- ...

1.3 FPL for Linear Losses

Algorithm 17 – FPL for linear losses

- 1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
- 2: Sample $n_0 \sim \mathcal{D}$. Let $\hat{x}_1 \in \arg \min_{x \in \mathcal{K}} \{-n_0^\top x\}$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict $\hat{x}_t$.
- 5: Observe the linear loss function, suffer loss $g_t^\top x_t$.
- 6: Update

$$\hat{x}_t = \arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} g_s^\top x + n_0^\top x \right\}$$

7: **end for**

- ...
- ...

1.4 FPL for Prediction from Expert Advice

Algorithm 18 – FPL for prediction from expert advice

- 1: Input: $\eta > 0$.
- 2: Draw n exponentially distributed variables $n(i) \sim e^{-\eta x}$.
- 3: Let $x_1 = \arg \min_{e_i \in \Delta_n} \{-e_i^\top n\}$.
- 4: **for** $t = 1$ to T **do**
- 5: Predict using expert i_t such that $\hat{x}_t = e_{i_t}$.
- 6: Observe the loss vector, suffer loss $g_t^\top \hat{x}_t = g_t(i_t)$.
- 7: Update (w.l.o.g. choose $\hat{x}_{t+1}$ to be a vertex)

$$\hat{x}_{t+1} = \arg \min_{x \in \Delta_n} \left\{ \sum_{s=1}^t g_s^\top x - n^\top x \right\}$$

8: **end for**

- Special case of Algorithm 17: the simplex Δ_n , costs in $[0, 1]$, and one-sided exponential noise $n(i) \sim e^{-\eta x}$ – the first use of perturbation in decision making (Hannan [1957]).
- The generic FPL bound (Corollary 5.9) is loose here, so Theorem 5.10 gives a dedicated analysis, tight up to constants.

1.5 Regret Bound for Expert-Advice FPL

Theorem (5.10)

Algorithm 18 outputs predictions $\hat{x}_1, \dots, \hat{x}_T \in \Delta_n$ with

$$(1 - \eta) \mathbb{E} \left[\sum_t g_t^\top \hat{x}_t \right] \leq \min_{x^* \in \Delta_n} \sum_t g_t^\top x^* + \frac{4 \log n}{\eta}.$$

Optimizing η . Choosing $\eta = \sqrt{(\log n)/T}$ gives

$$\text{Regret}_T = O(\sqrt{T \log n}),$$

matching the **Hedge** guarantee (Theorem 1.5) up to constants.

Why a separate analysis? — the order gap.

- **Generic FPL (Cor. 5.9)** on Δ_n : the uniform-noise instance has $\sigma \leq n^{1/4}$, $L \leq 1$, giving $\text{Regret}_T = O(DGn^{1/4}\sqrt{T})$ — a factor $n^{1/4}$ **worse** than the $O(\sqrt{T})$ of OGD (Theorem 3.1).
- **Expert-advice FPL (Thm 5.10)**, with exponential noise, removes that polynomial-in- n penalty: only a $\sqrt{\log n}$ dependence remains, on par with the standard multiplicative-weights / Hedge bound $O(\sqrt{T \log n})$.

1.6 Proof of Theorem 5.10 (1/2)

Proof. Set $g_0 = -n$. Applying **Lemma 5.4** (FTL-BTL) to $f_t(x) = g_t^\top x$,

$$\mathbb{E} \left[\sum_{t=0}^T g_t^\top u \right] \geq \mathbb{E} \left[\sum_{t=0}^T g_t^\top \hat{x}_{t+1} \right].$$

Therefore, telescoping and isolating the $t = 0$ term,

$$\begin{aligned} \mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - x^\star) \right] &\leq \mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \right] + \mathbb{E}[g_0^\top (x^\star - x_1)] \\ &\leq \mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \right] + \mathbb{E}[\|n\|_\infty \|x^\star - x_1\|_1] \\ &\leq \sum_{t=1}^T \mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] + \frac{4}{\eta} \log n. \quad (5.6) \end{aligned}$$

The 2nd line uses the **generalized Cauchy–Schwarz** inequality; the 3rd uses

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_\infty] \leq \frac{2 \log n}{\eta} \quad (\text{exercise}), \quad \|x^\star - x_1\|_1 \leq 2.$$

1.7 Proof of Theorem 5.10 (2/2)

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \leq 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \leq \|g_t\|_\infty \cdot \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] \leq \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $-n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $-n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned} \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] &= 1 - \Pr[-n(i_t) > v + g_t(i_t) \mid -n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^{\infty} \eta e^{-\eta x} dx}{\int_v^{\infty} \eta e^{-\eta x} dx} = 1 - e^{-\eta g_t(i_t)} \leq \eta g_t(i_t) = \eta g_t^\top \hat{x}_t. \end{aligned}$$

Substituting into (5.6),

$$\mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - x^*) \right] \leq \eta \sum_t \mathbb{E}_t [g_t^\top \hat{x}_t] + \frac{4 \log n}{\eta},$$

which rearranges to the theorem. □

1.7 Proof of Theorem 5.10 (2/2)

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \leq 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \leq \|g_t\|_\infty \cdot \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] \leq \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $-n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $-n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned} \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] &\leq 1 - \Pr[-n(i_t) > v + g_t(i_t) \mid -n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^{\infty} \eta e^{-\eta x} dx}{\int_v^{\infty} \eta e^{-\eta x} dx} = 1 - e^{-\eta g_t(i_t)} \leq \eta g_t(i_t) = \eta g_t^\top \hat{x}_t. \end{aligned}$$

My reading: this first step looks like it should really be “ $\leq$ ”, not “ $=$ ” – I sketch why on the next slide.

Substituting into (5.6),

$$\mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - x^*) \right] \leq \eta \sum_t \mathbb{E}_t [g_t^\top \hat{x}_t] + \frac{4 \log n}{\eta},$$

which rearranges to the theorem. □

1.8 The Marked Equality is Only “ $\leq$ ”: an $N = 2$ Check

The proof’s event is a **sufficient** condition for staying leader, not a necessary one: it ignores that competitors also pay losses.

Expert	Before g_t	After $g_{t(1)} = g_{t(2)} = 1$
1 = leader	S_1	$S_1 + 1$
2	$S_2 > S_1$	$S_2 + 1 > S_1 + 1$

Equal losses shift both scores by the same amount, so the arg min cannot change:

$$\Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t = e_1] = 0.$$

But the displayed equality would give

$$1 - e^{-\eta g_{t(1)}} = 1 - e^{-\eta} > 0.$$

Thus the marked step should be “ $\leq$ ”, not “ $=$ ”. The direction is still enough for the regret bound, so Theorem 5.10 is unaffected.

2. Adaptive Gradient Descent (AdaGrad)

2.1 Learning the Regularizer Online

Motivation. RFTL's regret bound (5.7) depends on the regularizer R ; the optimal R depends on both $\mathcal{K}$ and the cost sequence. So we learn it online — AdaGrad optimizes the regularization (line 5) to shrink the dominant gradient-norm term in (5.7).

Algorithm 19 — AdaGrad

- 1: Input: parameters $\eta, x_1 \in \mathcal{K}$.
- 2: Initialize: $G_0 = 0$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict x_t , suffer loss $f_t(x_t)$.
- 5: Update $G_t = G_{t-1} + \nabla_t \nabla_t^\top$ and define

$$\text{(diagonal)} \quad H_t = \arg \min_{H \succeq 0, H = \text{diag}(H)} \{G_t \cdot H^{-1} + \text{Tr}(H)\} = (\text{diag}(G_t))^{\frac{1}{2}}$$

$$\text{(full matrix)} \quad H_t = \arg \min_{H \succeq 0} \{G_t \cdot H^{-1} + \text{Tr}(H)\} = G_t^{\frac{1}{2}}$$

- 6: Update $y_{t+1} = x_t - \eta H_t^{-1} \nabla_t$, $x_{t+1} = \arg \min_{x \in \mathcal{K}} \|y_{t+1} - x\|_{H_t}^2$.
 - 7: **end for**
-

2.2 What Line 5 is Optimizing

AdaGrad chooses the matrix H_t that would have made the **observed** gradients small, with a trace penalty to keep the metric from blowing up:

$$H_t = \arg \min_H \{G_t \cdot H^{-1} + \text{Tr}(H)\}, \quad G_t = \sum_{s=1}^t \nabla_s \nabla_s^\top.$$

Definition (Two comparator classes)

$$\mathcal{H}_1 = \{H : H = \text{diag}(H), H \succeq 0, \text{Tr}(H) \leq 1\}, \quad \mathcal{H}_2 = \{H : H \succeq 0, \text{Tr}(H) \leq 1\}.$$

Lemma (5.11)

For $i \in \{1, 2\}$, with the corresponding final AdaGrad matrix H_T ,

$$\sqrt{\min_{H \in \mathcal{H}_i} \sum_{t=1}^T \|\nabla_t\|_H^{*2}} = \text{Tr}(H_T).$$

2.3 Regret Bound: Compete with the Best Fixed Metric

Theorem (5.12)

Let x_t be generated by Algorithm 19. With $\eta = \frac{D_\infty}{\sqrt{2}}$ for the diagonal version and $\eta = \frac{D}{\sqrt{2}}$ for the full-matrix version, for any $x^* \in \mathcal{K}$,

$$\text{Regret}_{T(\text{AdaGrad-diag})} \leq \sqrt{2}D_\infty \sqrt{\min_{H \in \mathcal{H}_1} \sum_t \|\nabla_t\|_H^{*2}},$$

$$\text{Regret}_{T(\text{AdaGrad-full})} \leq \sqrt{2}D \sqrt{\min_{H \in \mathcal{H}_2} \sum_t \|\nabla_t\|_H^{*2}}.$$

Reading. AdaGrad is nearly as good as the best fixed quadratic regularizer in hindsight, chosen from the diagonal or full PSD trace ball.

Why the theorem follows from Lemma 5.11.

The adaptive choice makes $\text{Tr}(H_T)$ exactly encode the best hindsight gradient norm over $\mathcal{H}_i$.

Why there is still a price.

The regularizer changes over time, so the proof needs one extra drift term involving $H_t - H_{t-1}$.

2.4 When AdaGrad Beats OGD

For the unit cube in $\mathbb{R}^d$, $D_\infty = 1$ while $D = \sqrt{d}$. The textbook compares the diagonal-AdaGrad and OGD bounds as

$$\text{Regret}_{T(\text{AdaGrad-diag})} \leq \sqrt{2 \text{Tr}\left((\text{diag}(G_T))^{\frac{1}{2}}\right)},$$

$$\text{Regret}_{T(\text{OGD})} \leq \sqrt{2d \text{Tr}(\text{diag}(G_T))}.$$

Sparse gradient geometry.

If only a few coordinates accumulate large gradient mass, AdaGrad can improve over OGD by up to a $\sqrt{d}$ factor.

Dense / Euclidean geometry.

If G_T is dense, or the feasible set is closer to a Euclidean ball, OGD can be better by the same order.

Takeaway. AdaGrad is not uniformly better than OGD; it wins when the data geometry is coordinate-sparse or otherwise mismatched to the Euclidean metric.

2.5 Analysis Roadmap for Theorem 5.12

The proof splits AdaGrad regret into two terms, then controls each by the same final quantity $\text{Tr}(H_T)$.

Lemma (5.13)

$$\text{Regret}_T \leq \frac{\eta}{2} (G_T \cdot H_T^{-1} + \text{Tr}(H_T)) + \frac{1}{2\eta} \sum_{t=0}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2.$$

Lemma 5.14. Gradient term:

$$G_T \cdot H_T^{-1} \leq \text{Tr}(H_T).$$

Lemma 5.15. Drift term:

$$\sum_t \|x_t - x^*\|_{H_t - H_{t-1}}^2 \leq D_*^2 \text{Tr}(H_T).$$

Choosing η balances the two terms and recovers Theorem 5.12.

2.6 Lemma 5.13: One-Step Identity

The update is a gradient step in the current metric:

$$y_{t+1} = x_t - \eta H_t^{-1} \nabla_t.$$

Subtract $x^\star$ and multiply by H_t :

$$\begin{aligned} y_{t+1} - x^\star &= x_t - x^\star - \eta H_t^{-1} \nabla_t, \\ H_t(y_{t+1} - x^\star) &= H_t(x_t - x^\star) - \eta \nabla_t. \end{aligned}$$

Multiplying the transpose of the first line by the second line gives the basic energy identity.

2.7 Lemma 5.13: Expanding the Square

Expanding in the H_t -norm gives the textbook equation (5.8):

$$\|y_{t+1} - x^\star\|_{H_t}^2 = \|x_t - x^\star\|_{H_t}^2 - 2\eta \nabla_t^\top (x_t - x^\star) + \eta^2 \nabla_t^\top H_t^{-1} \nabla_t. \quad (5.8)$$

Rearranging this identity will isolate the one-step regret term $\nabla_t^\top (x_t - x^\star)$.

2.8 Lemma 5.13: Projection Gives Descent

Since x_{t+1} is the projection of y_{t+1} in the norm induced by H_t ,

$$\|y_{t+1} - x^*\|_{H_t}^2 \geq \|x_{t+1} - x^*\|_{H_t}^2.$$

Combining this with (5.8):

$$\nabla_t^\top (x_t - x^*) \leq \frac{\eta}{2} \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} \left(\|x_t - x^*\|_{H_t}^2 - \|x_{t+1} - x^*\|_{H_t}^2 \right).$$

This is the OGD proof, except that the metric changes with t .

2.9 Lemma 5.13: Telescoping with Changing Metrics

Summing the one-step inequality from $t = 1$ to T :

$$\begin{aligned}\sum_{t=1}^T \nabla_t^\top (x_t - x^\star) &\leq \frac{\eta}{2} \sum_{t=1}^T \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} \|x_1 - x^\star\|_{H_0}^2 \\ &\quad + \frac{1}{2\eta} \sum_{t=1}^T \left(\|x_t - x^\star\|_{H_t}^2 - \|x_t - x^\star\|_{H_{t-1}}^2 \right) \\ &\quad - \frac{1}{2\eta} \|x_{T+1} - x^\star\|_{H_T}^2.\end{aligned}$$

With $H_0 = 0$ and dropping the final nonnegative term,

$$\sum_t \nabla_t^\top (x_t - x^\star) \leq \frac{\eta}{2} \sum_t \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} \sum_{t=0}^T \|x_t - x^\star\|_{H_t - H_{t-1}}^2.$$

2.10 Lemma 5.13: BTL Controls the Gradient Term

Define

$$\Psi_{t(H)} = \nabla_t \nabla_t^\top \cdot H^{-1}, \quad \Psi_0(H) = \text{Tr}(H).$$

By construction, H_t minimizes $\sum_{i=0}^t \Psi_i(H)$. Applying the BTL Lemma 5.4,

$$\begin{aligned} \sum_{t=1}^T \nabla_t^\top H_t^{-1} \nabla_t &= \sum_{t=1}^T \Psi_{t(H_t)} \\ &\leq \sum_{t=1}^T \Psi_{t(H_T)} + \Psi_0(H_T) - \Psi_0(H_0) \\ &= G_T \cdot H_T^{-1} + \text{Tr}(H_T). \end{aligned}$$

Substituting this into the previous slide proves Lemma 5.13.

2.11 Lemma 5.14: Explicit Optimizer

Proposition 5.16 solves the optimization in line 5. For $A \succeq 0$,

$$\arg \min_{X \succeq 0} \{A \cdot X^{-1} + \text{Tr}(X)\} = A^{\frac{1}{2}}.$$

With the diagonal constraint, the optimizer is

$$\arg \min_{X \succeq 0, X = \text{diag}(X)} \{A \cdot X^{-1} + \text{Tr}(X)\} = (\text{diag}(A))^{\frac{1}{2}}.$$

Applying this to $A = G_T$ gives

$$H_T = G_T^{\frac{1}{2}} \quad \text{or} \quad H_T = (\text{diag}(G_T))^{\frac{1}{2}}.$$

2.12 Lemma 5.14: Gradient Term is at Most Trace

For the full-matrix version,

$$G_T \cdot H_T^{-1} = G_T \cdot G_T^{-\frac{1}{2}} = \text{Tr}\left(G_T^{\frac{1}{2}}\right) = \text{Tr}(H_T).$$

For the diagonal version, H_T ignores off-diagonal entries:

$$G_T \cdot H_T^{-1} = \text{diag}(G_T) \cdot (\text{diag}(G_T))^{-\frac{1}{2}} = \text{Tr}\left((\text{diag}(G_T))^{\frac{1}{2}}\right) = \text{Tr}(H_T).$$

Therefore, in both cases,

$$G_T \cdot H_T^{-1} \leq \text{Tr}(H_T).$$

2.13 Lemma 5.15: Diagonal Drift Term

For diagonal AdaGrad, $G_t \succeq G_{t-1}$ implies $H_t - H_{t-1} \succeq 0$. Since for diagonal H , $x^\top H x \leq \|x\|_\infty^2 \text{Tr}(H)$,

$$\begin{aligned} \sum_{t=1}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2 &\leq \sum_{t=1}^T D_\infty^2 \text{Tr}(H_t - H_{t-1}) \\ &= D_\infty^2 \sum_{t=1}^T (\text{Tr}(H_t) - \text{Tr}(H_{t-1})) \\ &\leq D_\infty^2 \text{Tr}(H_T). \end{aligned}$$

The trace telescopes because the metric increments are PSD.

2.14 Lemma 5.15: Full-Matrix Drift Term

In the full-matrix case, $H_t - H_{t-1} \succeq 0$ and $\lambda_{\max}(A) \leq \text{Tr}(A)$ for $A \succeq 0$:

$$\begin{aligned} \sum_{t=1}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2 &\leq \sum_{t=1}^T D^2 \lambda_{\max}(H_t - H_{t-1}) \\ &\leq D^2 \sum_{t=1}^T \text{Tr}(H_t - H_{t-1}) \\ &= D^2 \sum_{t=1}^T (\text{Tr}(H_t) - \text{Tr}(H_{t-1})) \\ &\leq D^2 \text{Tr}(H_T). \end{aligned}$$

This proves Lemma 5.15.

2.15 Putting the Pieces Together

Lemma 5.13 plus Lemmas 5.14 and 5.15 gives

$$\begin{aligned}\text{Regret}_T &\leq \frac{\eta}{2}(2 \text{Tr}(H_T)) + \frac{D_*^2}{2\eta} \text{Tr}(H_T) \\ &= \left(\eta + \frac{D_*^2}{2\eta} \right) \text{Tr}(H_T),\end{aligned}$$

where $D_* = D_\infty$ for diagonal AdaGrad and $D_* = D$ for full AdaGrad.

Choosing $\eta = \frac{D_*}{\sqrt{2}}$ yields

$$\text{Regret}_T \leq \sqrt{2}D_* \text{Tr}(H_T).$$

Finally, Lemma 5.11 converts $\text{Tr}(H_T)$ back to the best hindsight metric:

$$\text{Tr}(H_T) = \sqrt{\min_{H \in \mathcal{H}_i} \sum_{t=1}^T \|\nabla_t\|_H^{*2}},$$

which is exactly Theorem 5.12.

3. Bibliographic Remarks

3.1 Notes & References

- **Origins.** Regularization for online learning: Grove et al. [2001]; Kivinen & Warmuth [2001].
- **Follow-the-(Perturbed-)Leader.** Kalai & Vempala [2005] coined *follow-the-leader* and analyzed random perturbation as a regularizer (FPL), reviving an idea of Hannan [1957].
- **Follow-the-Regularized-Leader.** *FTRL* coined by Shalev-Shwartz & Singer [2007]; identical *RFTL* by Abernethy et al. [2008]. Hazan & Kale [2008]: *RFTL* $\equiv$ Online Mirror Descent.
- **AdaGrad.** Duchi et al. [2010, 2011]; diagonal version in parallel by McMahan & Streeter [2010]. Analysis here follows Gupta et al. [2017].
- **Adaptive regularization in deep learning.** AdaGrad, RMSprop [Tieleman & Hinton 2012], Adam [Kingma & Ba 2014]; survey: Goodfellow et al. [2016].
- **Randomization $\leftrightarrow$ regularization.** In special cases, adding randomization $\approx$ deterministic strongly convex regularization [Abernethy et al. 2014, 2016].

4. Exercises

4.1 Exercise 1(b): Generalized Cauchy–Schwarz

Claim. For any norm $\|\cdot\|$ with dual norm $\|y\|_* := \sup_{\|z\| \leq 1} z^\top y$,

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

Proof. If $x = 0$ (or $y = 0$), both sides equal 0 and the inequality holds. So assume $x \neq 0$ from here on.

Since $x \neq 0$, the unit vector $u = x/\|x\|$ satisfies $\|u\| = 1$, hence it is admissible in the supremum defining the dual norm. Therefore

$$\|y\|_* = \sup_{\|z\| \leq 1} z^\top y \geq u^\top y = \frac{x^\top y}{\|x\|}.$$

Multiplying both sides by $\|x\| > 0$ yields

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

□

4.2 Theorem / Proof blocks

Theorem (Name of result)

State the claim here, with math like $R_T = O(\sqrt{T})$.

Proof. Sketch the argument; the QED square is added automatically.

□

4.3 Definition / Lemma blocks

Definition (Term)

f is α -strongly convex if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\alpha}{2} \|y - x\|^2.$$

Lemma

Use lemma the same way as theorem.

4.4 Two columns

Idea. Text on the left, matching math on the right.

- Bullet one
- Bullet two

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta \nabla f_t(x_t))$$
$$\Pi_{\mathcal{K}}(y) = \arg \min_{x \in \mathcal{K}} \|x - y\|$$

4.5 Incremental reveal

State the assumption.

4.5 Incremental reveal

State the assumption.

Then

4.5 Incremental reveal

State the assumption.

Then introduce the algorithm.

4.5 Incremental reveal

State the assumption.

Then introduce the algorithm.

Finally, conclude.

A big, centered takeaway statement.

4.6 Bibliography

Drop a `refs.bib` beside this file, then uncomment: